Model Evaluation and Performance Analysis

The Smart Classroom Energy System used a Logistic Regression model to predict whether motion was detected in the classroom. The model was trained using environmental sensor features, including light level, temperature, humidity, carbon dioxide concentration, and hour of the day.

The target variable, motion detected, was created from the original motion sensor data. A value of 0 represented No Motion, while a value of 1 represented Motion Detected. The original motion column was excluded from the model inputs to avoid target leakage.

The final model was evaluated on 13,152 test samples using a classification threshold of 0.724513. The model achieved an accuracy of 89.83%, a precision of 15.55%, a recall of 62.71%, and an F1-score of 24.92% for the Motion Detected class.

The confusion matrix showed that the model correctly classified 11,592 No Motion samples and 222 Motion Detected samples. It incorrectly predicted motion in 1,206 cases and missed 132 actual motion events.

The results indicate that the model performs well in identifying No Motion cases but has limited precision when predicting motion. The dataset is imbalanced, with substantially more No Motion samples than Motion Detected samples. Therefore, accuracy alone is insufficient to evaluate the model. Precision, recall, F1-score, and the confusion matrix provide a more complete understanding of its performance.

These results represent the model's performance on the available test dataset. They do not guarantee perfect performance in a real classroom environment.

Group Member Reflections

Group Member 1 Machine Learning Model and Data Analysis

During this project, I learned how to prepare and analyze a dataset using Kaggle Notebook. I gained practical experience in training the Logistic Regression model, selecting the classification threshold, and evaluating its performance using accuracy, precision, recall, F1-score, and a confusion matrix. The project helped me understand the challenges of working with imbalanced datasets and how machine learning can support intelligent classroom lighting. I also improved my Python programming and analytical skills.

Group Member 2 Dashboard, and Application Development

Through this project, I gained practical knowledge of developing a Smart Classroom Energy System that combines IoT concepts with machine learning. I learned how environmental sensor data could be used to support automated lighting decisions. I also contributed to the dashboard and application development, which helped me improve my programming, system integration, troubleshooting, and teamwork skills. This project demonstrated how technology could be used to improve energy efficiency in classrooms.